



PROG6212 POE

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RaceDay API Endpoint Plan

1. Introduction

The RaceDay API will provide the backend services required by the RaceDay event management system. The API will allow Organisers to manage road running, walking, and cycling events, while Participants can browse events, enrol in categories, and view their personal results.

The API will use RESTful HTTP methods to perform Create, Read, Update, and Delete (CRUD) operations. Role-based access control will be applied to ensure that Organisers and Participants can only perform actions appropriate to their roles.

2. User Roles

Organiser: Can create, update, and delete events, manage event categories, view enrolments for their events, and capture participant results.

Participant: Can create an account, browse events, select categories, enrol in events, view their own enrolments, and view their personal results.

3. HTTP Methods

- **GET** – Retrieves existing information.
- **POST** – Creates new information.
- **PUT** – Updates existing information.
- **DELETE** – Removes existing information.

4. API Endpoint Plan

4.1 Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new RaceDay user as an Organiser or Participant.	None	FirstName, LastName, Email, Password, Role	201 Created – account created. 400 Bad Request – invalid data. 409 Conflict – email already exists.
POST	/api/auth/login	Authenticate s a user and returns an access token.	None	Email, Password	200 OK – login successful and token returned. 401 Unauthorized – invalid credentials.

4.2 User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/profile	Retrieves the profile of the currently logged-in user.	Authenticated	None	200 OK – profile returned. 401 Unauthorized – authentication required
PUT	/api/users/profile	Updates the profile of the currently logged-in user.	Authenticated	PhoneNumber, City, EmergencyContact, ProfileImageUrl	200 OK – profile updated. 400 Bad Request – invalid data.

4.3 Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Retrieves all available RaceDay events	None	None	200 OK – list of events
GET	/api/events/{id}	Retrieves details of a specific event.	None	None	200 OK – event details. 404 Not Found – event does not exist.
POST	/api/events	Creates a new running, walking, or cycling event.	Organiser	EventName, Description, EventDate, Location, Distance, EventType	201 Created – event created. 400 Bad Request – invalid data
PUT	/api/events/{id}	Updates an existing event.	Organiser	EventName, Description, EventDate, Location, Distance, EventType	200 OK – event updated. 404 Not Found – event does not exist
DELETE	/api/events/{id}	Deletes an existing event.	Organiser	None	204 No Content – event deleted. 404 Not Found – event does not exist

4.4 Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Retrieves all categories available for an event.	Authenticated	None	200 OK – list of categories. 404 Not Found – event does not exist
GET	/api/categories/{id}	Retrieves details of a specific category.	Authenticated	None	200 OK – category details. 404 Not Found – category does not exist
POST	/api/events/{eventId}/categories	Creates a category for an event	Organiser	Category Name, Distance, EntryFee, MaximumParticipants	201 Created – category created. 404 Not Found – event does not exist
PUT	/api/categories/{id}	Updates an existing category.	Organiser	Category Name, Distance, EntryFee, MaximumParticipants	200 OK – category updated. 404 Not Found – category does not exist.
DELETE	/api/categories/{id}	Deletes an existing category.	Organiser	None	204 No Content – category deleted. 404 Not Found – category does not exist.

4.5 Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments	Enrols the logged-in participant into an event by selecting a category.	Participant	EventId, CategoryId	201 Created – enrolment created. 400 Bad Request – invalid event or category. 409 Conflict – participant already enrolled
GET	/api/enrolments/my	Retrieves all events the logged-in participant has entered.	Participant	None	200 OK – participant's enrolments
GET	/api/events/{eventId}/enrolments	Retrieves all participant enrolments for an organiser's event.	Organiser	None	200 OK – enrolment list. 403 Forbidden – organiser does not own event.
DELETE	/api/enrolments/{id}	Cancels the logged-in participant's enrolment.	Participant	None	204 No Content – enrolment cancelled. 404 Not Found – enrolment does not exist

4.6 Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/results	Records a participant's finish time and finishing position.	Organiser	EnrolmentId, FinishTime, Position, ResultStatus	201 Created – result recorded. 400 Bad Request – invalid data.
GET	/api/results/my	Retrieves the logged-in participant's personal results.	Participant	None	200 OK – participant's results.
GET	/api/events/{eventId}/results	Retrieves all participant results for an organiser's event.	Organiser	None	200 OK – event results. 403 Forbidden – organiser does not own event.
PUT	/api/results/{id}	Updates a participant's result	Organiser	FinishTime, Position, ResultStatus	200 OK – result updated. 404 Not Found – result does not exist.